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The Process of Methylmercury Accumulation in Rice (*Oryza sativa* L.)

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Supporting Information

ABSTRACT: Recent studies have shown that rice consumption can be an important pathway of methylmercury (MeHg) exposure to humans in Hg mining areas and also in certain inland areas of Southwestern China. The seed of rice has the highest ability to accumulate MeHg compared to other tissues. The main objective of this study was to investigate the process of (MeHg) accumulation in rice seed (*Oryza sativa* L.) by monitoring MeHg levels in specific tissues of rice plants experiencing various levels of Hg multisource pollution during a full rice growing season. Four groups of experimental plantations were utilized, distributed among a rural artisanal Hg production site and a regional background control site. Our results suggest that the newly deposited Hg is more readily transformed to MeHg and accumulated in rice plants than Hg forms with an extended residence time in soil, and soil is the potential source of MeHg in the tissues of rice plants. MeHg in soil was first absorbed by roots and then translocated to the above-ground parts (leaf and stalk). During the full rice growing season only a very small amount of MeHg was retained in the root section. In the premature plant, the majority of MeHg is located in the leaf and stalk; however, most of this MeHg was transferred to seed during the ripening period.



1. INTRODUCTION

The organic species methylmercury (MeHg) is of great concern due to its developmental neurotoxicity.[1] Because of its tendency to biomagnify in the food chain, MeHg concentrations in certain fish species can reach a level 10^6 times higher than ambient water Hg concentrations.[2] Consumption of fish, fish products, and marine mammals is currently considered as the main pathway of human exposure to MeHg, posing a worldwide human health threat.[3] However, recent studies have elucidated that rice consumption can be the main pathway of MeHg exposure to humans in Hg mining areas and also in certain inland areas in Southwestern China.[4−7] Generally, Hg concentrations in most foodstuffs (except for fish) are below 20 $\mu g\ kg^{-1}$ and mainly present in inorganic forms.[8,9] Regarding MeHg in rice, Horvat et al. [10] reported an elevated concentration of 140 $\mu g\ kg^{-1}$ in rice collected at the Wanshan Hg mining site, Guizhou province. Qiu et al.[7] reported that MeHg in rice grown at abandoned Hg mining areas contained levels >100 $\mu g\ kg^{-1}$ in the edible portion, which is 10−100 fold higher than that of other locally grown crop plants, such as corn (*Zea mays* L.), rape (*Brassica campestris*), tobacco (*Nicotiana tabacum*), and cabbage (*Brassica oleracea*). At Wuchuan, another large abandoned Hg mining area in NE Guizhou, similarly high levels of MeHg in rice plants have been reported.[11] More recently, Zhang et al.[5] determined that on average, the bioaccumulation factors (BAFs) for MeHg in rice were more than 800 times (with the maximum of 40 000 times) higher than those for inorganic Hg. Meng et al.[12] demonstrated that the rice seed (brown rice) has the highest ability to accumulate MeHg compared to other tissues, and that paddy soil could be a potential MeHg source to tissues of rice plants.

The rice paddy is one of the most prevalent land uses throughout South and East Asia, where rice provides the dominant staple food. The physicochemical conditions present in the paddy environment facilitate Hg-methylation due to the presence of a flora of sulfur-reducing bacteria (SRB).[13,14]

Gouxi as a rural artisanal Hg production site and Huaxi as a regional background control site were selected for this study. Detailed information concerning each sampling location is shown in the Supporting Information.

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Currently, available data show that rice seed has high accumulation potential for MeHg.[4,5,7,11,12] However, the dynamic process of MeHg translocation within rice plants is unknown. To date, the distribution of MeHg in rice plants during the rice growing season has not been investigated, and an understanding of the bioaccumulation pathways of MeHg in rice is still limited. To better understand the mechanisms and processes controlling MeHg accumulation in rice plants, we measured the MeHg concentration in tissues of rice plants collected from four designed experimental plots during one rice growing season.

2. MATERIALS AND METHODS

2.1. Experimental Design and Chemical Sampling Protocol. In order to investigate the process of MeHg accumulation in rice, four experimental plots of rice plantation were carefully designed (See Supporting Information for details). The first plot was a control site paddy (CS-P) consisting of a rice paddy field located at Huaxi (control site) receiving a comparatively low Hg input by atmospheric deposition. This field with low soil Hg concentration (THg: 0.29 ± 0.023 mg kg^{-1}) was irrigated with the local drinking water containing a low concentration of Hg (THg: 5.4 ± 1.9 ng L^{-1}). The second plot was a control site-box (CS-B) (dimension $1.5 \times 1.5 \times 0.4$ m), located at Huaxi, but filled with topsoil (0−10 cm) collected from a Hg-contaminated paddy field (THg: 30 ± 1.1 mg kg^{-1}) in Gouxi. This box received the same irrigation water as CS-P. The CS-B was situated beside the CS-P but was mounted on the roof about 2.5 m above the surrounding ground in order to avoid any contamination from soil particles by splashing during heavy rainfall. Both CS-P and CS-B received a low Hg input by atmospheric deposition throughout the growing season. The third plot was an artisanal site paddy (AS-P) located at Gouxi with a high Hg soil concentration (THg: 11 ± 3.4 mg kg^{-1}). This area receives an elevated Hg input through atmospheric deposition (both wet and dry deposition). The paddy was irrigated with streamwater contaminated with Hg (4200 ± 4900 ng L^{-1}). The fourth plot was an artisanal site-box (AS-B) with the same dimension as the CS-B and filled with topsoil (0−10 cm) collected from the Huaxi paddy field with a low level of Hg (THg: 0.44 ± 0.028 mg kg^{-1}). This box was placed adjacent to but above (mounted on a roof 2.5 m high) the surrounding ground in order to avoid any contamination from soil particles by splashing during heavy rainfall. The plants in the box were irrigated with the local drinking water containing low levels of Hg (14 ± 2.7 ng L^{-1}). Both AS-B and AS-P received a high atmospheric Hg depositional flux characteristic of the Gouxi area.

A rice cultivar (hybrid rice: Jinyou) widely grown throughout Guizhou province was selected for this experiment. The germinated seeds were cultivated in a paddy field at the control site for one month. Subsequently, the plants were transplanted into the soil of the four plots (with a density of 25 cm ×25 cm). The experimental plots were cultivated during the period June through September. In order to maintain flooded soil conditions, saturation to permanent immersion of the soil, with 2−4 cm of water above soil surface, was required during the growing period. The rice plants of the four plots were irrigated as required corresponding to the local weather during the rice growing season.

2.2. Sample Collection and Preparation. Five sampling campaigns were conducted during the rice growing season. Sampling was initiated one month after the plants were transplanted into the research plots (day 30), and thereafter every 15 days until the final harvest on day 90. For each experiment plot, two to six rice plants, as well as the corresponding soil from the root zone (10−20 cm depth), were collected during each sampling campaign. Samples of irrigation water and precipitation were collected concurrently using approved methodologies[15] (see Supporting Information for details). The corresponding soil samples were collected by hand using disposable polyethylene gloves. The rice plants were divided into three fractions: root, stalk, and leaf. Concerning the latter sampling campaigns (at day 75 and day 90) coinciding with rice ripening, rice grains were, in addition, manually separated using a scalpel in order to prevent rice grain loss.

Samples of rice tissue were cleaned in the field and the laboratory using drinking water, washed through immersion in deionized water in an ultrasonic bath upon return to the laboratory to remove deposited particles, and then air-dried in polyethylene bags to avoid cross contamination. The mass of each rice plant sample was recorded in the field after collection and then again in the laboratory after drying. The soil samples were sealed, double-bagged, stored in an ice-cooled container before being shipped to the laboratory within 24 h, and then stored in a refrigerator at −17 °C prior to being freeze-dried. All water samples were collected using ultraclean handling protocols, transferred into precleaned borosilicate glass bottles, and promptly acidified to 0.5% (v/v) using adequate volumes of concentrated hydrochloric acid. The bottled samples were then sealed, double-bagged, and transported to the laboratory within 24 h. Prior to Hg analysis, samples were stored in a refrigerator at +4 °C in the dark.

Rice plant tissues were ground to 150 mesh (IKA-A11 basic, IKA, Germany). Similarly, freeze-dried soil samples were homogenized to 150 mesh with a mortar before chemical analysis. Precautions were taken in order to avoid any cross-contamination during sample preparation. The grinder was thoroughly cleaned after processing each sample. Powdered samples were transferred into an open plastic dish and separately enclosed in polyethylene bags and then placed in a desiccator kept at +4 °C in the dark. The concentration of MeHg in soil, plant tissue, and water samples was analyzed. For soil and water samples, THg determination was performed as well.

2.3. Analytical Methods. The protocol for THg/MeHg analysis of soil and water samples, MeHg analysis of rice plant tissues, analysis of soil pH and organic matter, and statistical analysis of the resulting analytical data are described in the Supporting Information.

2.4. Quality Control. Quality control measurements consisted of method blanks, triplicates, matrix spikes, and the parallel analysis of several certified reference materials as described in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Total Hg and MeHg in Irrigation Water and Precipitation. CS-P, CS-B, and AS-B were irrigated by local drinking water with a low level of Hg, while AS-P was irrigated by streamwater with an elevated level of Hg (Supporting Information Table S1). The average THg concentration in the AS-P irrigation water (4200 ± 4900 ng L^{-1}) was significantly higher than that for AS-B (14 ± 2.7 ng L^{-1}) and that for CS-P and CS-B (5.4 ± 1.9 ng L^{-1}) (Supporting Information Table S2).

During the rice growing season, the THg concentration in precipitation at Gouxi was highly elevated (2900 ± 1400 ng L^{-1}) in comparison with Huaxi (27 ± 17 ng L^{-1}) (Table S2). Wang et al.[16] observed a significantly negative correlation between Hg flux from the soil surface and air Hg concentration in Hg mining areas. The average Hg dry deposition flux (0−9434 ng m^{-2} h^{-1}) in the Wanshan Hg mining area is reported to be 0−4 orders of magnitude higher than that in the background area.[16] In other Hg mining areas in Guizhou, such as Wuchuan, the dry deposition flux can reach up to 10916 ± 8339 ng m^{-2} h^{-1}.[17] Meng et al.[12] observed that the total gas mercury (TGM) concentration in ambient air at Gouxi, reaching up to 1650 ng m^{-3}, was much higher than that at Huaxi (6.2 ± 3.0 ng m^{-3}).[18] Li et al.[19] reported that the concentration of THg (up to 47 mg kg^{-1}) in topsoil (0−10 cm) in the Gouxi area was significantly elevated compared to sampling depths lower in the soil profile (0.2−1.0 mg kg^{-1}). These results indicate that both wet and dry deposition are responsible for Hg input to surface areas in Gouxi, whereas Huaxi as a control site has no direct point sources of Hg contamination.[12,18]

Previous studies have observed that the concentration of MeHg in air at Hg mining areas[20] and urban areas[21] is negligible. Our rough calculations show that the increased MeHg mass from wet deposition and irrigation water to soil at AS-B during the sampling season represents only 0.06−0.13% and 0.00034−0.00077% of the total increased MeHg mass in soil, respectively (see Supporting Information for details). We can therefore confirm that the relative importance of MeHg input from precipitation (Gouxi: 2.0 ± 0.15 ng L^{-1}; Huaxi: 0.28 ± 0.14 ng L^{-1}) and irrigation water (AS-P: 2.8 ± 1.5 ng L^{-1}; AS-B: 0.24 ± 0.02 ng L^{-1}; CS-P and CS-B: 0.13 ± 0.04 ng L^{-1}) to the plots was negligible (Table S2). Owing to the highly elevated Hg concentration in precipitation (wet and dry deposition) compared to that in the local drink water at Gouxi, the relative importance of airborne Hg input to the plot AS-B should be considered large. With respect to AS-P, Hg input from both precipitation and irrigation water would be considerable. For CS-P and CS-B however, the contributions of Hg from both precipitation and the irrigation water (the local drinking water) were trivial because of very low Hg concentration (precipitation: 27 ± 17 ng L^{-1}; local drinking water: 5.4 ± 1.9 ng L^{-1}).

3.2. Total Hg and MeHg in Rice Paddies. *3.2.1. Total Hg and MeHg in Rice Paddies before the Rice Plantation.* The average THg concentration in soil for AS-P and CS-B at planting was 11 ± 3.4 mg kg^{-1} and 30 ± 1.1 mg kg^{-1}, respectively, which was significantly higher than that of CS-P (0.29 ± 0.023 mg kg^{-1}) and that of AS-B (0.44 ± 0.028 mg kg^{-1}) (Table S2). The soil for CS-B was collected from a paddy (topsoil, 0−10 cm) impacted by artisanal Hg mining activities (150 m away). Li et al.[19] found that Hg in topsoil (0−10 cm) was primarily from deposition of atmospheric Hg emitted during local artisanal Hg mining. Total Hg concentrations in topsoil (0−10 cm) collected from the center of an artisanal Hg mining area can reach up to 21 mg kg^{-1}. However, the THg concentration in surface soil (0−10 cm) located about 1 km from the center of the artisanal Hg mining can be as low as 2.7 mg kg^{-1}.[19] The elevated Hg soil data recorded for the current study compares favorably with previously published data for Hg mining areas.[22−24] The lower soil THg concentration for AS-B and CS-P are within the range of 0.01−0.5 mg kg^{-1} representative of uncontaminated soils globally.[25] The highest initial MeHg concentration in soil was for plot AS-P (4.2 ± 0.23 μg kg^{-1}), with a downward tendency

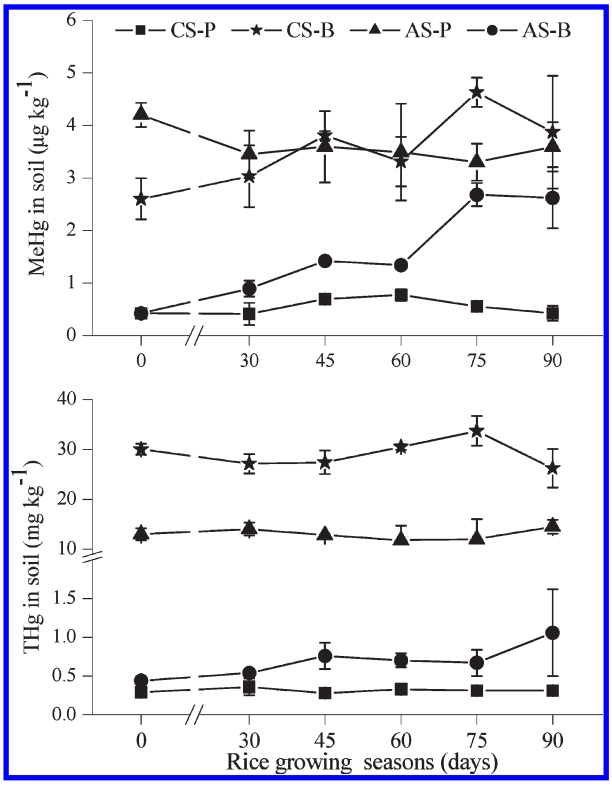


Figure 1. Concentration of THg and MeHg (dry weight with standard deviations) in soil collected from four experiment plots during the rice growing season.

from CS-B (2.6 ± 0.39 μg kg^{-1}) to CS-P (0.42 ± 0.10 μg kg^{-1}) and AS-B (0.42 ± 0.034 μg kg^{-1}) (Table S2).

3.2.2. Variation of Total Hg and MeHg in Rice Paddies during the Rice Growing Season. The variation of THg and MeHg concentrations in soil during the rice growing season is summarized in Figure 1 and Supporting Information Table S3 for the four experimental plots. For plots CS-B and AS-P, THg in soil was consistently above the domestic environmental quality standard for agricultural soil (1.5 mg kg^{-1})[26] during the sampling period. During the rice growing season, the mean concentration of MeHg in soil collected from AS-P (3.6 ± 0.49 μg kg^{-1}), AS-B (1.6 ± 0.99 μg kg^{-1}), and CS-B (3.6 ± 0.89 μg kg^{-1}) was significantly higher than that of CS-P (0.53 ± 0.20 μg kg^{-1}). The concentration of THg and MeHg in the soil of CS-P, CS-B, and AS-P showed a narrow variation with time. A more obvious temporal trend was present in the corresponding AS-B data (Figure 1 and Table S3). The concentration of THg (0.44 ± 0.028 mg kg^{-1}) and MeHg (0.42 ± 0.03 μg kg^{-1}) in soil from AS-B was low at the beginning of the rice growing season but increased by a factor of ~2.5 and ~6.2, respectively, when the rice was harvested (Figure 1). As described, air Hg deposition was the main source of Hg input to soil at AS-B, while the contribution of MeHg from both atmosphere input (wet/dry deposition) and irrigation water was considered negligible because of very low levels of MeHg. The enhancement of THg and MeHg in the soil at AS-B expressly implicates contribution from atmospheric deposition and in situ methylation. Historical large-scale Hg smelting activities combined with current artisanal Hg smelting has resulted in serious Hg contamination to the paddy soil of AS-P (THg: 13 ± 3.0 μg kg-1). Therefore, newly deposited Hg during the rice growing season cannot be

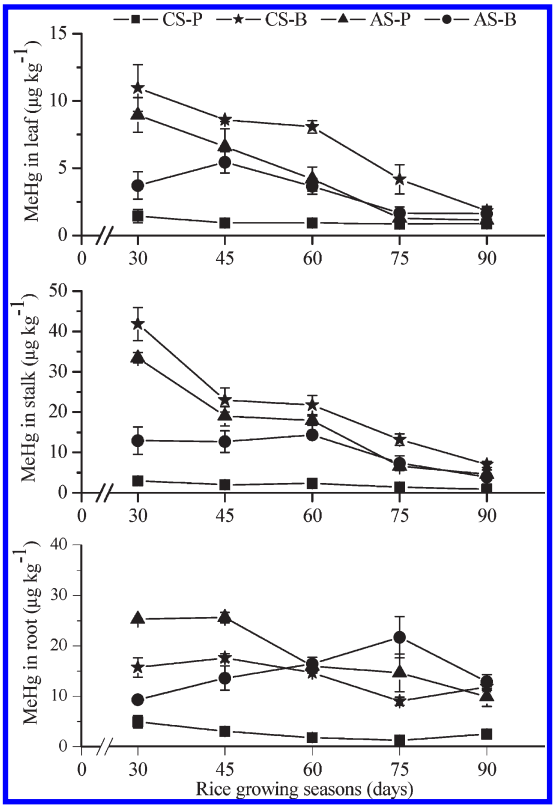


Figure 2. Concentration of MeHg (dry weight with standard deviations) in root, stalk, and leaf collected from four experiment plots during the rice growing season (rice plants before day 30 were unavailable).

distinguished from Hg forms at AS-P with an extended residence time in soil.

Wetlands are widely known as sites for MeHg production[27] and are generally considered as net sources of MeHg.[28] Paddies are submerged during the rice growing season and again flooded during the postharvest phase of fall and winter to speed up the decomposition of stalk. The ensuing anoxic conditions facilitate a flora of SRB.[13,14] Hence, rice paddy soil as a typical ephemeral wetland, has a high potential for Hg methylation, which results in the accumulation of MeHg in rice plants. Meng et al.[12] observed that the THg concentration in paddy soil at abandoned Hg mining areas can reach up to 131 ± 145 mg kg^{-1}. This is in direct contrast to the plot AS-B (THg: 1.1 ± 0.56 mg kg^{-1}), where the MeHg concentration was quite low (1.7 ± 0.7 $\mu g\ kg^{-1}$) compared to plot AS-B (2.6 ± 0.58 $\mu g\ kg^{-1}$). Even though THg concentrations in soil increased 2.5 times, MeHg concentrations in soil increased 6.2 times at AS-B, demonstrating that newly deposited Hg has a higher affinity to undergo methylation than the "old" mercury in soil. Harris et al.[29] also demonstrated that newly deposited Hg is more readily methylated and bioaccumulated to the food chain than "old" mercury in aquatic systems.[12,30] The net increase in Hg methylation in soil at AS-B after day 60 can be attributed to (1) sufficient Hg from precipitation when exposed to the high Hg deposition flux/air a long time, and (2) the high temperature during the period after day 60. However, the factors that control the methylation of Hg in paddy soil are likely to be complex. To better understand the mechanism of Hg methylation in paddy soil, further work is needed.

Table 1. Pearson's Correlation Matrix, Giving the Linear Correlation Coefficients (r) among the MeHg Levels in Leaf, Stalk, Root, and Soil during the Rice Growing Season (n = 58)

	root	stalk	leaf	soil
root	1			
stalk	0.73[a]	1		
leaf	0.73[a]	0.93[a]	1	
soil	0.62[a]	0.41[b]	0.36[b]	1

[a] Correlation is significant at the 0.001 level (two-tailed). [b] Correlation is significant at the 0.01 level (two-tailed). [c] Correlation is significant at the 0.05 level (two-tailed).

3.3. MeHg in Tissues of Rice Plant during the Rice Growing Season. The variations of MeHg concentration in rice plants collected from the experimental plots are shown in Figure 2 and Supporting Information Table S4. As implicitly stated in Section 2.1, the rice seedlings set out in the field experiments contained the same level of MeHg. However, throughout the rice growing season, the mean concentration of MeHg in the roots of rice plants from CS-B (16 ± 6.9 $\mu g\ kg^{-1}$), AS-P (13 ± 3.0 $\mu g\ kg^{-1}$), and AS-B (14 ± 4.2 $\mu g\ kg^{-1}$) was significantly higher than that in the roots of plants from CS-P (2.8 ± 1.5 $\mu g\ kg^{-1}$). Statistical analysis revealed a significantly positive correlation (r = 0.62, p < 0.001, n = 58) between root and soil MeHg (Table 1). Previous studies have observed that MeHg concentrations in roots were significantly correlated with the corresponding soil when the rice was harvested.[5,12] Schwesig et al.[31] have jointly reported that organic Hg in plants can be transported much more easily than inorganic Hg. Moreover, a recent study has found that phytochelatins, small peptides that detoxify heavy metals in plants, can sequester Hg^{2+} but not MeHg.[32] Therefore, the elevated MeHg concentrations in the roots of plants collected from AS-P, AS-B, and CS-B compared to those from CS-P is attributed to transport of MeHg from the corresponding paddy soil. We observed that the MeHg mass in roots reduced at day 60 as shown in Supporting Information Table S5. This could be attributed to MeHg translocation from roots to seeds (see Supporting Information for details).

Throughout the rice growing season, the mean concentration of MeHg in stalk and leaf at CS-B (stalk: 18 ± 13 $\mu g\ kg^{-1}$; leaf: 5.6 ± 3.7 $\mu g\ kg^{-1}$), AS-P (stalk: 13 ± 10 $\mu g\ kg^{-1}$; leaf: 3.5 ± 3.1 $\mu g\ kg^{-1}$) and AS-B (stalk: 8.8 ± 4.9 $\mu g\ kg^{-1}$; leaf: 2.9 ± 1.6 $\mu g\ kg^{-1}$) was significantly higher than those at CS-P (stalk: 1.9 ± 0.9 $\mu g\ kg^{-1}$; leaf: 1.0 ± 0.33 $\mu g\ kg^{-1}$) (K−S test, p < 0.001, both for stalk and leaf). The concentration of MeHg in stalk and leaf at CS-P, CS-B, AS-P, and AS-B reached the highest levels before the rice plant has grown for 60 days (Figure 2). There was a sharp decline in the concentration of MeHg in stalk and leaf tissues for all plots between day 60 and 75. This is attributed to the formation of seed. The lowest stalk (CS-P: 0.85 ± 0.18 $\mu g\ kg^{-1}$; CS-B: 7.0 ± 0.74 $\mu g\ kg^{-1}$; AS-P: 4.6 ± 1.1 $\mu g\ kg^{-1}$; AS-B: 3.9 ± 1.8 $\mu g\ kg^{-1}$) and leaf (CS-P: 0.87 ± 0.12 $\mu g\ kg^{-1}$; CS-B: 1.8 ± 0.33 $\mu g\ kg^{-1}$; AS-P: 1.1 ± 0.34 $\mu g\ kg^{-1}$; AS-B: 1.8 ± 0.33 $\mu g\ kg^{-1}$) MeHg concentration corresponded to the full ripening of rice seed on day 90 (Figure 2).

For all plots, rice seed recorded the highest concentration of MeHg compared to the other tissues (root, stalk, leaf, and hull) (Table S4). This is in agreement with previously reported data.[12] The mean concentration of MeHg in the ripe seeds followed the trend AS-P (46 ± 8.5 $\mu g\ kg^{-1}$) > CS-B (37 ± 5 $\mu g\ kg^{-1}$) > AS-B (28 ± 3.6 $\mu g\ kg^{-1}$) > CS-P (3.8 ± 0.46 $\mu g\ kg^{-1}$). Statistical

analysis shows that the concentration of MeHg in seed from AS-P, CS-B, and AS-B where the rice plant was exposed to an elevated level of soil MeHg was significantly higher than that from CS-P (K−S test, $p < 0.001$).

Generally, THg in food crops is below a concentration of 20 μg kg^{-1}, defined as the maximum permissible limit recommended by Chinese National Standard Agency[33] and is present in the less toxic form of inorganic Hg.[8] Organic forms of Hg are of greater concern due to higher toxicity and potentially severe effects on humans.[1] The average absorption rate of MeHg by the human body is about 95%.[34] Ripe rice seed obtained from the Hg source-modulated experiments (CS-B, AS-P, and AS-B) exceeded the recommended THg limit solely on the basis of THg concentration, posing a potential threat to the health of local residents.[4,6,7]

Although the soil used for plots CS-P and AS-B was of the same original provenance, the MeHg concentration in plants from AS-B was significantly higher than that from CS-P (K−S test, $p < 0.001$) (Tables S2 and S4), and this is attributed to the elevated MeHg levels in soil. Nevertheless, rice plants at CS-B experiencing low Hg levels in air and precipitation exhibited higher tissue MeHg concentration than for CS-P, which demonstrates the potential for increased MeHg production as a function of the background concentration of THg in the soil.

Significant and positive correlations were observed among the MeHg concentrations in root, stalk, leaf, hull, seed, and the corresponding soil (Table 1) during the rice growing season,[12] which implies that the source of MeHg in the tissues of rice plant should be unique. Principal component analysis (PCA) was performed for our data set (Table 2); one principal component (PC) with eigenvalue >1 was extracted. This PC had a high loading in all of the parameters. Our interpretation of this result is that the concentration of MeHg in paddy soil is the only factor that controls the concentration of MeHg in the tissues of rice plant, and that paddy soil is the major source of MeHg to rice plant, a statement which is in agreement with the previous observation.[12] However, we cannot rule out the possibility that MeHg could be formed in situ in rice tissues. More study is needed to elucidate the contribution of MeHg in rice from different pathways.

Table 2. Principal Component Factor Loadings and Percent Variance of the Data Explained by Each Factor from Concentrations of MeHg in Stalk, Leaf, Root, and Soil during the Rice Growing Season

rice tissue or source of MeHg	F1-PCR	F2-PCR
stalk	0.92	−0.31
leaf	0.91	−0.36
root	0.91	0.14
soil	0.66	0.73
% of variance explained	73%	19%
cumulative of total variance	92%	

3.4. MeHg Mass in Tissues of Rice Plant during the Rice Growing Season. The mass of MeHg (ng plant^{-1}) in the tissues of rice plants collected from each plot during the rice growing season are shown in Figure 3 and Table S5.

After a steady increase, the maximum mass of MeHg in stalk and leaf occurred at day 60 (Figure 3 and Table S5). Our study demonstrates that MeHg formed in soil could readily cross root

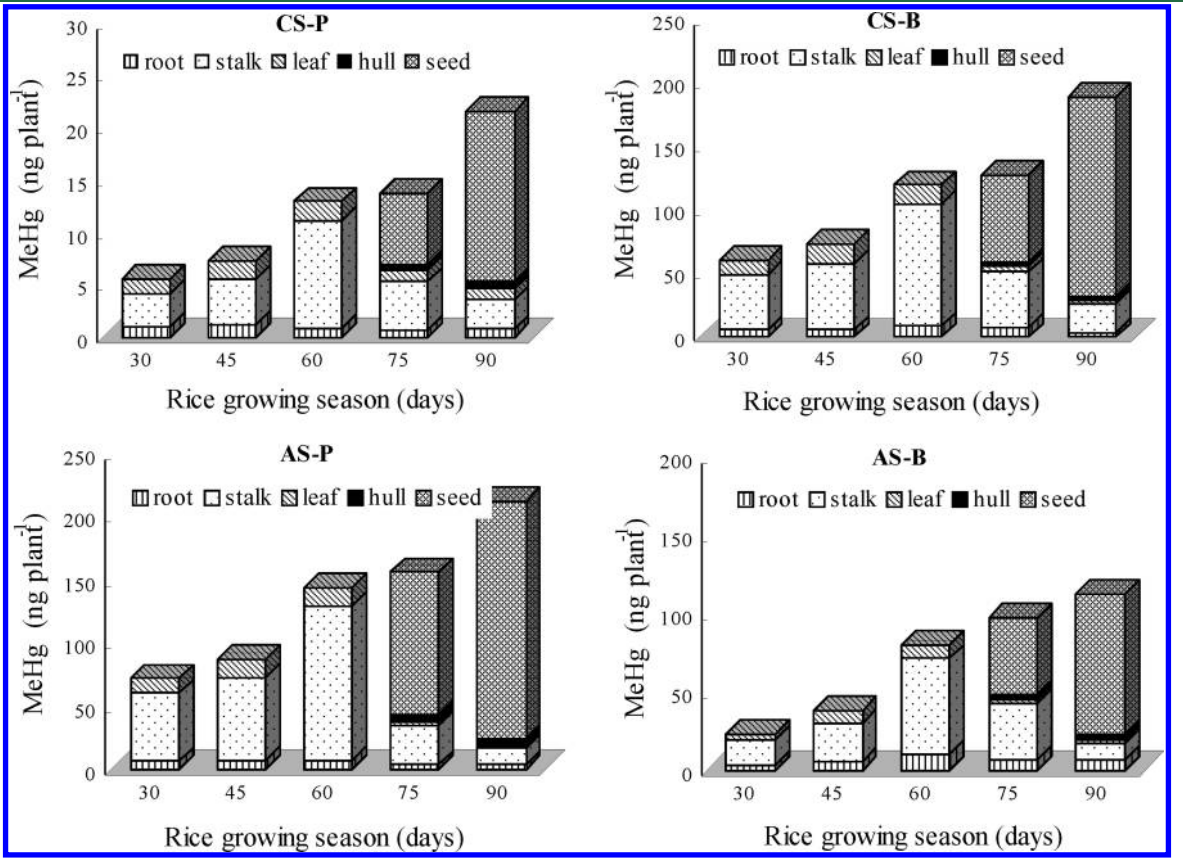


Figure 3. MeHg mass in tissues of rice plants collected from four experiment plots during the rice growing season.

barriers and bioaccumulate in the above-ground part of rice plants. A recent study showed that MeHg in rice seeds is present almost exclusively as CH_3Hg-L-cysteinate (CH_3HgCys), a complex that is thought to be responsible for the transfer of MeHg across the blood−brain and placental barriers.[35] It is well-known that proteins are formed and also bioaccumulate in the above ground part of the rice plant before rice seeds start to form and then are transported to seeds as the rice ripens.[36,37] An ongoing complexation between plant-associated MeHg and peptides to give, for example, CH_3HgCys, is thus implied during the process of rice ripenes.

When the rice grains start to ripen (day 60), plant growth begins to focus primarily on increasing the biomass of rice grain, whereas the biomass of rice stalk and leaf increases more slowly. Both the concentration of MeHg and the mass of MeHg in stalk and leaf were decreased sharply when the seed started to form (days 60), reaching a minimum level at rice harvest (Figures 2 and 3, and Tables S4 and S5).

Nutrient analysis of whole grains during the ripening period showed that more than 90% of K and 50% of P (in the form of phytic acid) found at maturity had accumulated in the grains.[38] Electron microprobe X-ray analysis indicated that Mg, P, and K were rapidly translocated into the grains during the ripening period and thence were highly concentrated in the grains when harvested.[38] The pattern of codeposition in the grain may suggest that most MeHg is exported to the grain during the ripening period.[38] This information and the results in this work confirm the transfer of MeHg from stalk and leaf to seed.

The pattern of bioaccumulation and translocation of MeHg from root to the above-ground tissues of rice plants appears identical for the four experiment plots. After rice plants are transplanted, MeHg in soil is first absorbed by the plant roots. MeHg in roots is then combined with protein, polysaccharide, and nucleic acid.[39] In the premature plant, the majority of the plant's MeHg is located in the leaf and stalk, but this is transferred to seed during the ripening period (Figures 2 and 3, and Tables S4 and S5).

The persistent increasing MeHg concentration in the soil collected from plot AS-B during the rice growing season suggests that newly deposited Hg is more readily transformed to MeHg and accumulated in rice plant than Hg forms with an extended residence time in soil. Elevated MeHg concentrations were observed in the tissues of rice cultivated in plots CS-B, AS-P, and AS-B (exposed to elevated MeHg soil) compared to plants cultivated in CS-P (exposed to relatively low MeHg soil). This observation is attributed to the higher concentration of MeHg in the soil directly (AS-P, AS-B) or indirectly (CS-B) affected by artisanal mining activities, suggesting that soil is the potential source of MeHg to the tissues of rice plants. During the rice growing season, this MeHg is absorbed by plant roots and behaves as a mobile plant nutrient, translocating from older parts of the plant to grain as the rice seed begins to develop.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information. Description of sampling location; detailed experimental design; method of precipitation collection; protocol for analysis of THg/MeHg analysis in soil and water, MeHg in rice plant tissue, soil pH, and organic matter; statistical analysis of analytical data; the QC/QA protocol; probable reason for the reduction of MeHg in roots; a rough calculation on the contribution of MeHg coming from wet precipitation and irrigation to plot AS-B; five tables; one figure. This material is available free of charge via the Internet at http://pubs.acs.org.

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